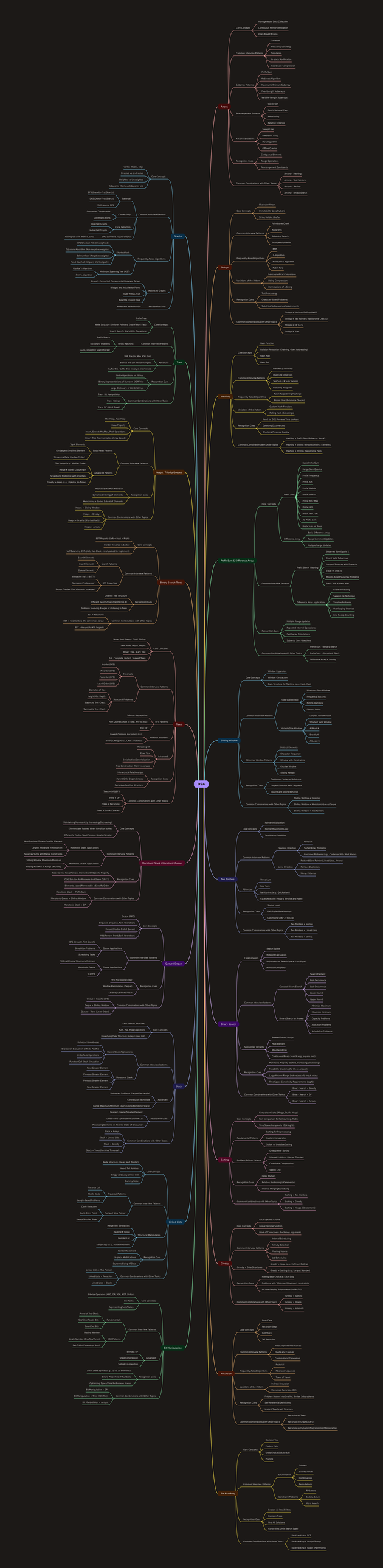

- **DSA**
  - **Arrays**
    - Core Concepts
      - Homogeneous Data Collection
      - Contiguous Memory Allocation
      - Index-Based Access
    - Common Interview Patterns
      - Traversal
      - Frequency Counting
      - Simulation
      - In-place Modification
      - Coordinate Compression
    - Subarray Patterns
      - Prefix Sum
      - Kadane's Algorithm
      - Maximum/Minimum Subarray
      - Fixed-Length Subarrays
      - Variable-Length Subarrays
    - Rearrangement Patterns
      - Cyclic Sort
      - Dutch National Flag
      - Partitioning
      - Relative Ordering
    - Advanced Patterns
      - Sweep Line
      - Difference Array
      - Mo's Algorithm
      - Offline Queries
    - Recognition Cues
      - Contiguous Elements
      - Range Operations
      - Rearrangement Constraints
    - Common Combinations with Other Topics
      - Arrays + Hashing
      - Arrays + Two Pointers
      - Arrays + Sorting
      - Arrays + Binary Search
  - **Strings**
    - Core Concepts
      - Character Arrays
      - Immutability (Java/Python)
      - String Builder / Buffer
    - Common Interview Patterns
      - Palindrome Check
      - Anagrams
      - Substring Search
      - String Manipulation
    - Frequently Asked Algorithms
      - KMP
      - Z-Algorithm
      - Manacher's Algorithm
      - Rabin-Karp
    - Variations of the Pattern
      - Lexicographical Comparison
      - String Compression
      - Permutations of a String
    - Recognition Cues
      - Text Processing
      - Character-Based Problems
      - Substring/Subsequence Requirements
    - Common Combinations with Other Topics
      - Strings + Hashing (Rolling Hash)
      - Strings + Two Pointers (Palindrome Checks)
      - Strings + DP (LCS)
      - Strings + Tries
  - **Hashing**
    - Core Concepts
      - Hash Function
      - Collision Resolution (Chaining, Open Addressing)
      - Hash Map
      - Hash Set
    - Common Interview Patterns
      - Frequency Counting
      - Duplicate Detection
      - Two Sum / K Sum Variants
      - Grouping Anagrams
    - Frequently Asked Algorithms
      - Rabin-Karp (String Hashing)
      - Bloom Filter (Existence Checks)
    - Variations of the Pattern
      - Custom Hash Functions
      - Rolling Hash (Substrings)
    - Recognition Cues
      - Need for O(1) Average Time Lookups
      - Counting Occurrences
      - Checking Presence Quickly
    - Common Combinations with Other Topics
      - Hashing + Prefix Sum (Subarray Sum K)
      - Hashing + Sliding Window (Distinct Elements)
      - Hashing + Strings (Palindrome Pairs)
  - **Prefix Sum & Difference Array**
    - Core Concepts
      - Prefix Sum
        - Basic Prefix Sum
        - Range Sum Queries
        - Prefix Frequency
        - Prefix XOR
        - Prefix Modulo
        - Prefix Product
        - Prefix Min / Max
        - Prefix GCD
        - Prefix AND / OR
        - 2D Prefix Sum
        - Prefix Sum on Trees
      - Difference Array
        - Basic Difference Array
        - Range Increment Updates
        - Multiple Range Updates
    - Common Interview Patterns
      - Prefix Sum + Hashing
        - Subarray Sum Equals K
        - Count Valid Subarrays
        - Longest Subarray with Property
        - Equal 0s and 1s
        - Modulo-Based Subarray Problems
        - Prefix XOR + Hash Map
      - Difference Array Applications
        - Event Processing
        - Sweep Line Technique
        - Timeline Problems
        - Overlapping Intervals
        - Line Sweep Counting
    - Recognition Cues
      - Multiple Range Updates
      - Repeated Interval Operations
      - Fast Range Calculations
      - Subarray Sum Questions
    - Common Combinations with Other Topics
      - Prefix Sum + Binary Search
      - Prefix Sum + Monotonic Stack
      - Difference Array + Sorting
  - **Sliding Window**
    - Core Concepts
      - Window Expansion
      - Window Contraction
      - Data Structure for Tracking (e.g., Hash Map)
    - Common Interview Patterns
      - Fixed Size Window
        - Maximum Sum Window
        - Frequency Tracking
        - Rolling Statistics
        - Distinct Count
      - Variable Size Window
        - Longest Valid Window
        - Shortest Valid Window
        - At Most K
        - Exactly K
        - At Least K
    - Advanced Window Patterns
      - Distinct Elements
      - Character Frequency
      - Window with Constraints
      - Circular Window
      - Sliding Median
    - Recognition Cues
      - Contiguous Subarray/Substring
      - Longest/Shortest Valid Segment
      - Expand and Shrink Behavior
    - Common Combinations with Other Topics
      - Sliding Window + Hashing
      - Sliding Window + Monotonic Queue/Deque
      - Sliding Window + Two Pointers
  - **Two Pointers**
    - Core Concepts
      - Pointer Initialization
      - Pointer Movement Logic
      - Termination Condition
    - Common Interview Patterns
      - Opposite Direction
        - Pair Sum
        - Sorted Array Problems
        - Container Problems (e.g., Container With Most Water)
      - Same Direction
        - Fast and Slow Pointer (Linked Lists, Arrays)
        - Remove Duplicates
        - Merge Patterns
    - Advanced
      - Three Sum
      - Four Sum
      - Partitioning (e.g., Quickselect)
      - Cycle Detection (Floyd's Tortoise and Hare)
    - Recognition Cues
      - Sorted Input
      - Pair/Triplet Relationships
      - Optimizing O(N^2) to O(N)
    - Common Combinations with Other Topics
      - Two Pointers + Sorting
      - Two Pointers + Linked Lists
      - Two Pointers + Strings
  - **Binary Search**
    - Core Concepts
      - Search Space
      - Midpoint Calculation
      - Adjustment of Search Space (Left/Right)
      - Monotonic Property
    - Common Interview Patterns
      - Classical Binary Search
        - Search Element
        - First Occurrence
        - Last Occurrence
        - Lower Bound
        - Upper Bound
      - Binary Search on Answer
        - Minimize Maximum
        - Maximize Minimum
        - Capacity Problems
        - Allocation Problems
        - Scheduling Problems
    - Specialized Variants
      - Rotated Sorted Arrays
      - Peak Element
      - Mountain Array
      - Continuous Binary Search (e.g., square root)
    - Recognition Cues
      - Monotonic Property (Sorted, Increasing/Decreasing)
      - Feasibility Checking (for BS on Answer)
      - Large Answer Range (not necessarily input array)
      - Time/Space Complexity Requirements (log N)
    - Common Combinations with Other Topics
      - Binary Search + Greedy
      - Binary Search + DP
      - Binary Search + Arrays
  - **Sorting**
    - Core Concepts
      - Comparison Sorts (Merge, Quick, Heap)
      - Non-Comparison Sorts (Counting, Radix)
      - Time/Space Complexity (O(N log N))
    - Fundamental Patterns
      - Sorting for Preprocessing
      - Custom Comparator
      - Stable vs Unstable Sorting
    - Problem-Solving Patterns
      - Greedy After Sorting
      - Interval Problems (Merge, Overlap)
      - Coordinate Compression
      - Sweep Line
    - Recognition Cues
      - Order Matters
      - Relative Positioning (of elements)
      - Interval Merging/Scheduling
    - Common Combinations with Other Topics
      - Sorting + Two Pointers
      - Sorting + Greedy
      - Sorting + Heaps (Kth element)
  - **Greedy**
    - Core Concepts
      - Local Optimal Choice
      - Global Optimal Solution
      - Proof of Correctness (Exchange Argument)
    - Common Interview Patterns
      - Interval Scheduling
      - Activity Selection
      - Meeting Rooms
      - Job Scheduling
    - Greedy + Data Structures
      - Greedy + Heap (e.g., Huffman Coding)
      - Greedy + Sorting (e.g., Largest Number)
    - Recognition Cues
      - Making Best Choice at Each Step
      - Problems with "Minimum/Maximum" constraints
      - No Overlapping Subproblems (unlike DP)
    - Common Combinations with Other Topics
      - Greedy + Sorting
      - Greedy + Heaps
      - Greedy + Intervals
  - **Recursion**
    - Core Concepts
      - Base Case
      - Recursive Step
      - Call Stack
      - Tail Recursion
    - Common Interview Patterns
      - Tree/Graph Traversal (DFS)
      - Divide and Conquer
      - Combinatorial Generation
    - Frequently Asked Algorithms
      - Factorial
      - Fibonacci Sequence
      - Tower of Hanoi
    - Variations of the Pattern
      - Indirect Recursion
      - Memoized Recursion (DP)
    - Recognition Cues
      - Problem Broken into Smaller, Similar Subproblems
      - Self-Referential Definitions
      - Implicit Tree/Graph Structure
    - Common Combinations with Other Topics
      - Recursion + Trees
      - Recursion + Graphs (DFS)
      - Recursion + Dynamic Programming (Memoization)
  - **Backtracking**
    - Core Concepts
      - Decision Tree
      - Explore Path
      - Undo Choice (Backtrack)
      - Pruning
    - Common Interview Patterns
      - Enumeration
        - Subsets
        - Subsequences
        - Combinations
        - Permutations
      - Constraint Problems
        - N-Queens
        - Sudoku Solver
        - Word Search
    - Recognition Cues
      - Explore All Possibilities
      - Decision Trees
      - Find All Solutions
      - Constraints Limit Search Space
    - Common Combinations with Other Topics
      - Backtracking + DFS
      - Backtracking + Arrays/Strings
      - Backtracking + Graph (Pathfinding)
  - **Graphs**
    - Core Concepts
      - Vertex (Node), Edge
      - Directed vs Undirected
      - Weighted vs Unweighted
      - Adjacency Matrix vs Adjacency List
    - Common Interview Patterns
      - Traversal
        - BFS (Breadth-First Search)
        - DFS (Depth-First Search)
        - Multi-source BFS
      - Connectivity
        - Connected Components
        - DSU Applications
      - Cycle Detection
        - Directed Graphs
        - Undirected Graphs
      - DAG (Directed Acyclic Graph)
        - Topological Sort (Kahn's, DFS)
    - Frequently Asked Algorithms
      - Shortest Path
        - BFS Shortest Path (Unweighted)
        - Dijkstra's Algorithm (Non-negative weights)
        - Bellman-Ford (Negative weights)
        - Floyd-Warshall (All-pairs shortest path)
      - Minimum Spanning Tree (MST)
        - Kruskal's Algorithm
        - Prim's Algorithm
    - Advanced Graphs
      - Strongly Connected Components (Kosaraju, Tarjan)
      - Bridges and Articulation Points
      - Euler Path/Circuit
      - Bipartite Graph Check
    - Recognition Cues
      - Nodes and Relationships
  - **Tries**
    - Core Concepts
      - Prefix Tree
      - Node Structure (Children Pointers, End-of-Word Flag)
      - Insert, Search, StartsWith Operations
    - Common Interview Patterns
      - String Matching
        - Prefix Search
        - Dictionary Problems
        - Auto-complete / Spell Checker
    - Advanced
      - XOR Trie (for Max XOR Pair)
      - Bitwise Trie (for Integer ranges)
      - Suffix Trie / Suffix Tree (rarely in interviews)
    - Recognition Cues
      - Prefix Operations on Strings
      - Binary Representations of Numbers (XOR Trie)
      - Large Dictionary of Words/Strings
    - Common Combinations with Other Topics
      - Trie + Bit Manipulation
      - Trie + Strings
      - Trie + DP (Word Break)
  - **Heaps / Priority Queues**
    - Core Concepts
      - Min-Heap, Max-Heap
      - Heap Property
      - Insert, Extract-Min/Max, Peek Operations
      - Binary Tree Representation (Array-based)
    - Common Interview Patterns
      - Basic Heap Patterns
        - Top K Elements
        - Kth Largest/Smallest Element
        - Streaming Data (Median Finder)
      - Advanced Patterns
        - Two Heaps (e.g., Median Finder)
        - Merge K Sorted Lists/Arrays
        - Scheduling Problems (with priorities)
        - Greedy + Heap (e.g., Dijkstra, Huffman)
    - Recognition Cues
      - Repeated Min/Max Retrieval
      - Dynamic Ordering of Elements
      - Maintaining a Sorted Subset of Elements
    - Common Combinations with Other Topics
      - Heaps + Sliding Window
      - Heaps + Greedy
      - Heaps + Graphs (Shortest Path)
      - Heaps + Arrays
  - **Binary Search Trees**
    - Core Concepts
      - BST Property (Left < Root < Right)
      - Inorder Traversal is Sorted
      - Self-Balancing BSTs (AVL, Red-Black - rarely asked to implement)
    - Common Interview Patterns
      - Search Patterns
        - Search Element
        - Insert Element
        - Delete Element
      - BST Properties
        - Validation (Is it a BST?)
        - Successor/Predecessor
        - Range Queries (Find elements in range)
    - Recognition Cues
      - Ordered Tree Structure
      - Efficient Search/Insert/Delete (log N)
      - Problems Involving Ranges or Ordering in Trees
    - Common Combinations with Other Topics
      - BST + Recursion
      - BST + Two Pointers (for conversion to LL)
      - BST + Heaps (for Kth largest)
  - **Trees**
    - Core Concepts
      - Node, Root, Parent, Child, Sibling
      - Leaf Node, Depth, Height
      - Binary Tree, N-ary Tree
      - Full, Complete, Perfect, Skewed Trees
    - Common Interview Patterns
      - Traversals
        - Inorder (DFS)
        - Preorder (DFS)
        - Postorder (DFS)
        - Level Order (BFS)
      - Structural Problems
        - Diameter of Tree
        - Height/Max Depth
        - Balanced Tree Check
        - Symmetric Tree Check
    - DFS Patterns
      - Subtree Aggregation
      - Path Queries (Root to Leaf, Any-to-Any)
      - Tree DP
    - Ancestor Problems
      - Lowest Common Ancestor (LCA)
      - Binary Lifting (for LCA, Kth Ancestor)
    - Advanced
      - Rerooting DP
      - Euler Tour
      - Serialization/Deserialization
      - Tree Construction (from traversals)
    - Recognition Cues
      - Hierarchical Relationships
      - Parent-Child Dependencies
      - Recursive/Iterative Structure
    - Common Combinations with Other Topics
      - Trees + DFS/BFS
      - Trees + DP
      - Trees + Recursion
      - Trees + Stacks/Queues
  - **Monotonic Stack / Monotonic Queue**
    - Core Concepts
      - Maintaining Monotonicity (Increasing/Decreasing)
      - Elements are Popped When Condition is Met
      - Efficiently Finding Next/Previous Greater/Smaller
    - Common Interview Patterns
      - Monotonic Stack Applications
        - Next/Previous Greater/Smaller Element
        - Largest Rectangle in Histogram
        - Subarray Sums with Range Constraints
      - Monotonic Queue Applications
        - Sliding Window Maximum/Minimum
        - Finding Max/Min in Range Efficiently
    - Recognition Cues
      - Need to Find Next/Previous Element with Specific Property
      - O(N) Solution for Problems that Seem O(N^2)
      - Elements Added/Removed in a Specific Order
    - Common Combinations with Other Topics
      - Monotonic Stack + Prefix Sum
      - Monotonic Queue + Sliding Window
      - Monotonic Stack + DP
  - **Queue / Deque**
    - Core Concepts
      - Queue (FIFO)
      - Enqueue, Dequeue, Peek Operations
      - Deque (Double-Ended Queue)
      - Add/Remove Front/Back Operations
    - Common Interview Patterns
      - Queue Applications
        - BFS (Breadth-First Search)
        - Simulation Problems
        - Scheduling Tasks
      - Deque Applications
        - Sliding Window Maximum/Minimum
        - Monotonic Queue
        - 0-1 BFS
    - Recognition Cues
      - FIFO Processing Order
      - Window Maintenance (Deque)
      - Level-by-Level Traversal
    - Common Combinations with Other Topics
      - Queue + Graphs (BFS)
      - Deque + Sliding Window
      - Queue + Trees (Level Order)
  - **Stack**
    - Core Concepts
      - LIFO (Last-In, First-Out)
      - Push, Pop, Peek Operations
      - Underlying Data Structure (Array/Linked List)
    - Common Interview Patterns
      - Classic Stack Applications
        - Balanced Parentheses
        - Expression Evaluation (Infix to Postfix)
        - Undo/Redo Operations
        - Function Call Stack Simulation
      - Monotonic Stack
        - Next Greater Element
        - Previous Greater Element
        - Previous Smaller Element
        - Next Smaller Element
    - Advanced
      - Histogram Problems (Largest Rectangle)
      - Contribution Technique
      - Range Maximum/Minimum Query (using Monotonic Stack)
    - Recognition Cues
      - Nearest Greater/Smaller Element
      - Linear-Time Optimization (from N^2)
      - Processing Elements in Reverse Order of Encounter
    - Common Combinations with Other Topics
      - Stack + Arrays
      - Stack + Linked Lists
      - Stack + Greedy
      - Stack + Trees (Iterative Traversal)
  - **Linked Lists**
    - Core Concepts
      - Node Structure (Value, Next Pointer)
      - Head, Tail Pointers
      - Singly vs Doubly Linked List
      - Dummy Node
    - Common Interview Patterns
      - Traversal Patterns
        - Reverse List
        - Middle Node
        - Length-Based Problems
      - Fast and Slow Pointer
        - Cycle Detection
        - Cycle Entry Point
        - Happy Number Style
    - Structural Manipulation
      - Merge Two Sorted Lists
      - Reverse K Group
      - Reorder List
      - Deep Copy (e.g., Random Pointer)
    - Recognition Cues
      - Pointer Movement
      - In-place Modifications
      - Dynamic Sizing of Data
    - Common Combinations with Other Topics
      - Linked Lists + Two Pointers
      - Linked Lists + Recursion
      - Linked Lists + Stacks
  - **Bit Manipulation**
    - Core Concepts
      - Bitwise Operators (AND, OR, XOR, NOT, Shifts)
      - Bit Masks
      - Representing Sets/States
    - Common Interview Patterns
      - Fundamentals
        - Power of Two Check
        - Set/Clear/Toggle Bits
        - Count Set Bits
      - XOR Patterns
        - Missing Number
        - Single Number (One/Two/Three)
        - Pair Tricks (Swapping, Sum)
    - Advanced
      - Bitmask DP
      - State Compression
      - Subset Enumeration
    - Recognition Cues
      - Small State Spaces (e.g., up to 20 elements)
      - Binary Properties of Numbers
      - Optimizing Space/Time for Boolean States
    - Common Combinations with Other Topics
      - Bit Manipulation + DP
      - Bit Manipulation + Tries (XOR Trie)
      - Bit Manipulation + Arrays